File

Name: OPT3-3-K02
Description: KISSsoft example
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	12.500	
Speed (1/min)	[n]	2950.0	782.7
Number of load cycles (in mio.)	[NL]	3540.000	939.184
Torque (Nm)	[T]	40.5	152.5
Application factor	[KA]	1.25	
Required service life (h)	[H]	20000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	128.529	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	2.0000	
Transverse module (mm)	[mt]	2.0612	
Normal diametral pitch (1/in)	[Pnd]	12.70000	
Transverse diametral pitch (1/in)	[Ptd]	12.32276	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	14.0000	
Number of teeth	[z]	26	98
Facewidth (mm)	[b]	27.00	26.00
Hand of gear		left	right
Accuracy grade	[Q-DIN 3961:1978]	6	6
Inner diameter (mm)	[di]	0.00	182.94
Shaft diameter (mm)	[ds]	0.00	146.39
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1
31CrMoV9, Nitriding steel, gas-nitrided, ISO 6336-5 Figure 13a/14a (MQ)

Gear 2
42CrMo4 (3), Through hardened steel, nitrided, ISO 6336-5 Figure 13b/14b (MQ)

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HV 800	HV 550
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	420.00	363.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1250.00	998.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1100.00	1100.00
Yield point (N/mm²)	[σS]	900.00	900.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	3.00	3.00
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	20.00	20.00
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	Mobilgear 600 XP 150	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	150.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	14.70
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	75.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	27.396
(mm)	[a0.e/i]	27.366 /27.330
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.814 / 0.816

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000

Ramp angle	[αKP]	0.000	
	not topping		
Manufacturing center distance (mm)	[a0]	101.148	
(mm)	[a0.e/i]	101.001 /100.910	
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.788 / 0.791	

4.1.1 Information on final machining

		------- Gear 1 ----------	Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000

Type of profile modification:	for high load capacity gearboxes		
Tip relief (µm)	[Ca L/R]	5.0 / 5.0	5.0 / 5.0

4.2 Basic data

Overall transmission ratio	[itot]	-3.769
Gear ratio	[u]	3.769
Transverse module (mm)	[mt]	2.061
Transverse pressure angle (°)	[αt]	20.562
Working pressure angle (°)	[αwt]	21.416
(°)	[αwt.e/i]	21.439 /21.393
Working pressure angle at normal section (°)	[αwn]	20.829
Base helix angle (°)	[βb]	13.140
Helix angle at operating pitch circle (°)	[βw]	14.077
Reference center distance (mm)	[ad]	127.796
Sum of profile shift coefficients	[Σxi]	0.3739

		------- Gear 1 ----------	Gear 2 -------
Profile shift coefficient	[x]	0.3000	0.0739
Generating profile shift coefficient	[xE.e/i]	0.2588/ 0.2416	0.0189/-0.0154
Virtual gear no. of teeth	[zn]	28.256	106.504
Involute length (mm)	[l_dFa-l_dFf]	4.073	4.133

4.3 Diameters and their allowances

		------- Gear 1 ----------	Gear 2 -------
Reference diameter (mm)	[d]	53.592	202.000
Base diameter (mm)	[db]	50.178	189.132
Tip alteration (mm)	[k*mn]	-0.015	-0.015
Tip diameter (mm)	[da]	58.762	206.266
(mm)	[da.e/i]	58.762 /58.752	206.266 /206.256
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.010	0.000 / -0.010
Tip form diameter (mm)	[dFa]	58.762	206.266
(mm)	[dFa.e/i]	58.762 /58.752	206.266 /206.256
Active tip diameter (mm)	[dNa]	58.762	206.266
(mm)	[dNa.e/i]	58.762 /58.752	206.266 /206.256
V-Circle diameter (mm)	[dv]	54.792	202.296
(mm)	[dv.e/i]	54.627 /54.558	202.076 /201.939

Operating pitch diameter (mm)	[dw]	53.899	203.159
(mm)	[dw.e/i]	53.908 /53.891	203.191 /203.127
Root diameter (mm)	[df]	49.792	197.296
(mm)	[df.e/i]	49.627 /49.558	197.076 /196.939
Active root diameter (mm)	[dNf]	51.490	199.438
(mm)	[dNf.e/i]	51.521 /51.466	199.478 /199.403
Root form diameter (mm)	[dFf]	51.338	198.542
(mm)	[dFf.e/i]	51.240 /51.201	198.352 /198.235

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.500	0.500
Effective tip clearance (mm)	[c.e/i]	0.704 / 0.590	0.642 / 0.563
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.160 / 0.113	0.622 / 0.525
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	2.585	2.133
(mm)	[ha.e/i]	2.585 / 2.580	2.133 / 2.128
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	1.900	2.352
(mm)	[hf.e/i]	1.982 / 2.017	2.462 / 2.531
Tooth height (mm)	[h]	4.485	4.485

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	34.918 /34.896	24.935 /24.927
Roll angle to dNf (°)	[ξNf.e/i]	13.345 /13.066	19.209 /19.137
Roll angle at dFf (°)	[ξFf.e/i]	11.854 /11.630	18.108 /17.989

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.7892	1.6246
Normal tooth thickness at tip circle (mm)	[san]	1.301	1.619
(mm)	[san.e/i]	1.241 / 1.208	1.542 / 1.486
Normal tooth thickness at tip form circle (mm)	[sFan]	1.301	1.619
(mm)	[sFan.e/i]	1.241 / 1.208	1.542 / 1.486
Normal space width at root circle (mm)	[efn]	0.000	1.444
(mm)	[efn.e/i]	0.000 / 0.000	1.458 / 1.467
Pitch on reference circle (mm)	[pt]	6.476	6.476
Base pitch (mm)	[pbt]	6.063	6.063
Transverse pitch on contact-path (mm)	[pet]	6.063	6.063
Lead height (mm)	[pz]	675.271	2545.252
Axial pitch (mm)	[px]	25.972	25.972

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	2.130	1.589
Specific sliding at the tip	[ζa]	0.451	0.471

Specific sliding at the root	[ζf]	-0.890	-0.821
Mean specific sliding	[ζm]	0.460	
Sliding factor on tip	[Kga]	0.256	0.191
Sliding factor on root	[Kgf]	-0.191	-0.256

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	41.879
Transverse contact ratio	[εα]	1.569
	[εα.e/m/i]	1.578 / 1.567/ 1.556
Overlap ratio	[εβ]	1.001
Total contact ratio	[εγ]	2.570
	[εγ.e/m/i]	2.579 / 2.568/ 2.558
Length of path of contact (mm)	[gα]	9.514
(mm)	[gα.e/m/i]	9.569 / 9.503/ 9.437
Approach path of contact (mm)	[gf]	4.064
(mm)	[gf.e/m/i]	4.107 / 4.058/ 4.008
Recess path of contact (mm)	[ga]	5.450
(mm)	[ga.e/m/i]	5.461 / 5.445/ 5.429

		Gear 1	Gear 2
Addendum contact ratio	[ε]	0.899	0.670
	[ε.e/i]	0.901 / 0.895	0.677 / 0.661
Length T1-A and T2-A (mm)	[T1A,T2A]	5.776	41.155
(mm)	[.e/i]	5.721 / 5.843	41.155 / 41.142
Length T1-B and T2-B (mm)	[T1B,T2B]	9.227	37.704
(mm)	[.e/i]	9.227 / 9.217	37.649 / 37.768
Length T1-C and T2-C (mm)	[T1C,T2C]	9.840	37.090
(mm)	[.e/i]	9.829 / 9.852	37.047 / 37.134
Length T1-D and T2-D (mm)	[T1D,T2D]	11.839	35.092
(mm)	[.e/i]	11.784 / 11.906	35.092 / 35.079
Length T1-E and T2-E (mm)	[T1E,T2E]	15.290	31.641
(mm)	[.e/i]	15.290 / 15.280	31.586 / 31.705
Length T1-T2 (mm)	[T1T2]	46.931	
(mm)	[.e/i]	46.876 / 46.985	
Diameter of single contact point B (mm)	[d-B]	53.464	203.610
(mm)	[d-B.e/i]	53.464 / 53.457	203.570 /203.658
Diameter of single contact point D (mm)	[d-D]	55.484	201.734
(mm)	[d-D.e/i]	55.437 / 55.542	201.734 /201.725

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	1510.05	
Axial force (N)	[Fa]	376.50	376.50
Radial force (N)	[Fr]	566.44	566.44
Normal force (N)	[Fnorm]	1656.15	1656.15
Nominal circumferential force per mm (N/mm)	[w]	58.08	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	1501.43	

Axial force (N)	[Faw]	376.50		376.50
Radial force (N)	[Frw]		588.89	
Circumferential speed reference circle (m/s)	[v]		8.28	
Circumferential speed operating pitch circle (m/s)	[v(dw)]		8.33	

5.2 Contact stiffness

Running-in value (µm)	[yp]		0.6	
Running-in value (µm)	[yf]		0.4	
Tolerances f_{pe}, f_f and $f_{H\beta}$, matching the tolerances in section 7				
Correction factor	[CM]		0.800	
Gear blank factor	[CR, bs/b, sr/mn]		0.931	(0.500,3.500)
Basic rack factor	[CB]		0.975	
Material coefficient	[E/Est]		1.000	
Single stiffness (N/mm/µm)	[c']		12.465	
Theoretical single stiffness (N/mm/µm)	[c'th]		19.161	
Contact stiffness (N/mm/µm)	[cγα]		17.786	
Contact stiffness (N/mm/µm)	[cγβ]		15.118	
Reduced mass (kg/mm)	[mRed]		0.00862	
Resonance speed (min-1)	[nE1]		16681	
Resonance ratio (-)	[N]		0.177	
Subcritical range				

5.3 Calculation of K factors

Running-in value (µm)	[yα]		0.3	
Bearing distance l of pinion shaft (mm)	[l]		42.000	
Distance s of pinion shaft (mm)	[s]		4.200	
Outside diameter of pinion shaft (mm)	[dsh]		21.000	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]			4
0:a), 1:b), 2:c), 3:d), 4:e)				
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']			-1.00
Without stiffening				
Effective flank line deviation (µm)	[Fβy]		4.97	
from deformation of shaft (µm)	[fsh·B1]		0.49	
fsh (µm)	[fsh]		0.98	
B1	[B1]		0.50	
fHβ5 (µm)	[fHβ5]		6.50	
Flank line:	crowned [$C_\beta = 0.5 \cdot (f_{m\alpha}+f_{sh})$]			
Position of contact pattern:	favorable			
from production tolerances (µm)	[fmα*B2]		6.36	
B_2=0.50				
Tooth trace deviation, theoretical (µm)	[Fβx]		5.85	
Running-in value (µm)	[yβ]		0.88	

5.4 K factors

Dynamic factor	[Kv]		1.131
Face load factors			
- Flank	[KHβ]		1.458
- Tooth root	[KFβ]		1.368
- Scuffing	[KBβ]		1.458
Transverse load factors			
- Flank	[KHα]		1.143
- Tooth root	[KFα]		1.143
- Scuffing	[KBα]		1.143

Application factor	[KA]	1.250	
Mesh load factor	[Kγ]	1.000	

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1 ----------	Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.2416	-0.0154
Tooth form factor	[YF]	0.972	1.071
Stress correction factor	[YS]	2.131	2.096
Load application angle (°)	[αFen]	21.119	20.287
Load distribution influence factor	[fε]	0.777	
Load application diameter, virtual spur gear (mm)	[d_{en}]	58.132	214.385
Load application diameter (mm)	[d_{en}]	55.211	203.377
Bending moment arm (mm)	[hFe]	1.947	2.352
(-)	[hFe*]	0.974	1.176
Tooth root thickness at critical cross section (mm)	[sFn]	4.307	4.521
(-)	[sFn*]	2.153	2.261
Root fillet radius at critical cross section (mm)	[ρF]	0.959	0.931
(-)	[ρF*]	0.479	0.466
Tangent contact point, virtual spur gear (mm)	[x,y]	2.153 /25.056	
Tangent contact point, virtual spur gear (mm)	[x,y]		2.261 /98.844
Diameter at critical cross section (mm)	[d_{sFn}]	50.298	197.740
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	2.246	2.427
Helix angle factor	[Yβ]	0.967	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	27.00	26.00
Nominal stress at tooth root (N/mm²)	[σF0]	56.03	63.05
Tooth root stress (N/mm²)	[σF]	123.87	139.40
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.989	0.997
Surface factor	[YRrelT]	0.990	0.990
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	0.868	0.891
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.850	0.880
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	840.00	726.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	509.77	456.12
Limit strength tooth root (N/mm²)	[σFG]	713.67	638.56

6.1 Safety factors

		------- Gear 1 ----------	Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	5.76	4.58
Transmittable power (kW)	[kWRating]	51.44	40.90

7 Calculation of flank strength (pitting)

		------- Gear 1 ----------	Gear 2 -------
Zone factor	[ZH]	2.380	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.798	

Helix angle factor	[Zβ]	1.015	
Effective facewidth (mm)	[beff]	26.00	
Nominal contact stress (N/mm²)	[σH0]	428.72	
Contact stress at operating pitch circle (N/mm²)	[σHw]	658.00	
Coefficient [fZCa] 1.20 (Helical gear sets without flank modifications)			
Single tooth contact factor	[ZB,ZD]	1.10	1.10
Contact stress (N/mm²)	[σHB, σHD]	720.80	720.80
Lubricant coefficient for N_L	[ZL]	0.988	0.988
Speed factor at N_L	[ZV]	0.991	0.991
Roughness factor for N_L	[ZR]	0.788	0.788
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.867	0.889
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.669	0.686
Limited pitting is permissible: No			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	836.40	684.90
Pitting stress limit (N/mm²)	[σHG]	836.40	684.90

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.27	1.04
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.16	0.95
Safety regarding transmittable torque	[SHBD²]	1.35	0.90
Transmittable power (kW)	[kWRating]	16.83	11.29

8 Micropitting

Calculation method according to	ISO/TS 6336-22:2018	
Lubricant load according to FVA Info sheet 54/7 10, Mobilgear 600 XP 150		
Reference data FZG-C Test:		
Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	128.037
Contact temperature (°C)	[θBRef,A]	253.806
Lubrication gap thickness (µm)	[hRef,A]	0.043
Specific film thickness in test	[λGFT]	0.085

		------- Gear 1	---------- Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	1.000	
Permissible specific film thickness	[λGFP]	0.120	
Interim results in accordance with ISO/TS 6336-22:2018			
Coefficient of friction	[μ_m]	0.086	
Lubricant factor	[XL]	1.000	
Roughness factor	[XR]	1.722	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tooth mass temperature (°C)	[θM]	77.114	
Tip relief factor	[XCa(A)]	1.789	
Loss factor	[HV]	0.108	
Equivalent Young's modulus (N/mm²)	[Er]	226373.626	
Pressure-viscosity coefficient (m²/N)	[α38]	0.02051	
Dynamic viscosity (Ns/m²)	[ηtM]	24.871	
Roughness average value (µm)	[Ra]	3.000	

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes		1		1
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)			1	
Diameter (mm)	[dy]	52.373		204.913
Relative radius of curvature (mm)	[ρred]		6.472	
Load sharing factor	[XY]		0.640	
X_Y interpolated between the values $X_Y(\varepsilon_b=0.8)$ and $X_Y(\varepsilon_b=1.2)$ according to ISO/TC60/WG6.				
Contact stress (N/mm²)	[pH]		470.100	
Contact stress (N/mm²)	[pdyn]		721.513	
Minimal specific film thickness	[λGFY]		0.089	(h_Y=0.268 µm)
Safety against micropitting	[Sλ(B)]		0.747	

9 Scuffing load capacity

Calculation method according to		ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	5.00	5.00
Optimal tip relief (µm)	[Ceff]	4.08	
C_a taken as optimal in the calculation. 0=no, 1=yes		1	1

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.235	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	12
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.500	
Effective facewidth (mm)	[beff]	26.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]	136.813	
$K_{B\gamma}$ = 1.235, $w_{Bt} \cdot K_{B\gamma}$ = 169.021			
Angle factor	[Xαβ]	0.993	
ε_1 =0.899, ε_2 =0.670			

9.1 Flash temperature-criteria

		---------- Pair ----------		
Lubricant factor	[XL]	0.848		
Tooth mass temperature (°C)	[θMi]	78.96		
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$				
Average flash temperature (Δ°C)	[θflm]	8.42		
Scuffing temperature (°C)	[θS]	440.38		
Γ coordinates (point of highest temperature)	[Γ]	0.378		
[Γ.A]= -0.413, [Γ.E]= 0.554				
Maximum flash temperature (Δ°C)	[θflamax]	15.66		
Maximum contact temperature (°C)	[θBmax]	94.62		
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058		
Approach factor	[XJ]	1.000		
Load sharing factor	[XΓ]	0.641		
Dynamic viscosity (mPa*s)	[ηM]	26.82	(75.0	°C)
Coefficient of friction	[μ_m]	0.073		
Required safety	[SBmin]	2.000		
Margin of safety for scuffing, flash temperature	[SB]	18.626		

9.2 Integral temperature-criteria

		---------- Pair ----------

Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	78.88	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	441.52	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.246	
Dynamic viscosity (mPa*s)	[ηOil]	26.82	(75.0 °C)
Mean coefficient of friction	[μ_m]	0.084	
Geometry factor	[XBE]	0.241	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.789	
Mean flash temperature (Δ°C)	[θflaint]	5.55	
Integral tooth flank temperature (°C)	[θint]	87.21	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	5.063	
Safety factor for transmitted moment (int.-T.)	[SSL]	30.021	

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 d24	DIN 3967 d25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.060 / -0.085	-0.080 / -0.130

10.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	12.000
Base tangent length (no backlash) (mm)	[Wk]	21.869	70.992
Base tangent length with allowance (mm)	[Wk.e/i]	21.813 /21.789	70.917 /70.870
(mm)	[ΔWk.e/i]	-0.056 / -0.080	-0.075 / -0.122
Diameter of measuring circle (mm)	[dMWk.m]	54.484	201.338

10.3 Measurement over balls and pins

		------- Gear 1 ----------	Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	3.608	3.371
Effective diameter of ball/pin (mm)	[DMeff]	3.750	3.500
Radial single-ball measurement, no backlash (mm)	[MrK]	30.180	103.670
Radial single-ball measurement (mm)	[MrK.e/i]	30.117 /30.091	103.566 /103.501
Diameter of measuring circle (mm)	[dMMr.m]	54.880	202.331
Diametral measurement over two balls, no backlash (mm)	[MdK]	60.360	207.340
Diametral measurement over two balls (mm)	[MdK.e/i]	60.234 /60.181	207.132 /207.002
Diametral measurement over pins, no backlash (mm)	[MdR]	60.360	207.340
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	60.234 /60.181	207.132 /207.002
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	60.234 /60.181	207.132 /207.002

10.4 Tooth thickness

		------- Gear 1 ----------	Gear 2 -------
Medium tip diameter (mm)	[da.m]	58.757	206.261
Reference chordal height from da.m (mm)	[hac]	2.639	2.143
Tooth thickness at height hac, chord (mm)	[sc]	3.576	3.249
(mm)	[sc.e/i]	3.517 / 3.493	3.169 / 3.120
Tooth thickness on reference circle, arc (mm)	[sn]	3.578	3.249

(mm)	[sn.e/i]	3.518 / 3.493	3.169 / 3.119

10.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.016 / -0.016
Radial backlash (mm)	[jrw.e/i]	0.306 / 0.166
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.239 / 0.129
Normal backlash (mm)	[jnw.e/i]	0.216 / 0.118
Center distance allowances (mm)	[Aa.e/i]	0.020 / -0.020
Backlash free center distance (mm)	[aControl.e/i]	128.343 / 128.243
Backlash free center distance, allowances (mm)	[jta]	-0.186 / -0.286

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	51.159	198.942
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.041	0.295
Tip clearance (mm)	[c0.i(aControl)]	0.324	0.297

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.1345/ 0.0730	(0°8'4"/ 0°4'23")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.5071/ 0.2752	(0°30'26"/ 0°16'31")

11 Gear tolerances

		Gear 1	Gear 2
According to DIN 3961:1978			
Accuracy grade	[Q-DIN3961]	6	6
Single pitch deviation (µm)	[fp]	±7.00	±8.00
Normal base pitch deviation (µm)	[fpe]	±7.00	±8.00
Adjacent pitch difference (µm)	[fu]	9.00	10.00
Total cumulative pitch deviation (µm)	[Fp]	25.00	29.00
Pitch span deviation (µm)	[Fpz/8]	±16.00	±18.00
Profile form deviation (µm)	[ff]	6.00	6.00
Profile slope deviation (µm)	[fHα]	±5.00	±5.00
Total profile deviation (µm)	[Ff]	8.00	8.00
Helix form deviation (µm)	[ffβ]	5.50	5.50
Helix slope deviation (µm)	[fHβ]	±9.00	±9.00
Total helix deviation (µm)	[Fβ]	10.00	10.00
Runout (µm)	[Fr]	16.00	19.00
Tooth Thickness Variation (µm)	[Rs]	10.00	11.00
Single flank composite, total (µm)	[Fi']	26.00	30.00
Single flank composite, tooth-to-tooth (µm)	[fi']	11.00	11.00
Radial composite, total (µm)	[Fi'']	20.00	24.00
Radial composite, tooth-to-tooth (µm)	[fi'']	8.00	10.00
According DIN 58405:1972 (Precision Mechanics)			
Axis alignment error (µm)	[fp]	±21.85	±21.85
Flank direction error (µm)	[fβ]	±5.67	±5.67
Runout (µm)	[Trk, Fr]	24.00	30.00
Composite error (µm)	[Fi'']	25.00	32.00
Tooth-to-tooth composite error (µm)	[fi'']	9.00	11.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99		
Total helix deviation for calculation (µm)	[Fβ]	10.00
Distance between bearings for calculation (mm)	[LG]	42.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 8.08
Maximum value for axis inclination error (µm)	[fΣδ]	± 16.15

12 Modifications and determination of the tooth form

12.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, linear

Caa = 5.000 µm LCa = 0.865 ·mn dCa = 57.038 mm

Gear 2

Symmetric (both flanks)

- Tip relief, linear

Caa = 5.000 µm LCa = 0.865 ·mn dCa = 204.910 mm

12.1.1 Tip relief verification

		------- Gear 1 ---------- Gear 2 -------	
Diameter (mm)	[dcheck]	58.712	206.216
Tip relief left/right (µm)	[Ca L/R]	4.9 / 4.9	4.8 / 4.8
Diameter (mm)	[dNa.i]	58.752	206.256
Tip relief left/right (µm)	[Ca L/R]	5.0 / 5.0	5.0 / 5.0

12.2 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	129.829

13.1 Masses, stiffnesses and moments of inertia

		------- Gear 1 ---------- Gear 2 -------	
Mass (kg)	[m]	0.489	1.159
Total mass (kg)	[Σm]	1.648	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.0001801	0.01075
System (da+df)/2...di (kg*m²)	[J]	0.0009366	

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	0.246
Torsion when subjected to nominal torque (°)	[δcr]	0.009
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	62.866

13.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.086
Wear sliding coefficient by Niemann	[ζw]	0.721
Loss factor	[HV]	0.108
Gear power loss (kW)	[PVZ]	0.116
Meshing efficiency (%)	[ηz]	99.068

13.4 Indications for the manufacturing by wire cutting

		------- Gear 1	---------- Gear 2 -------
Deviation from theoretical tooth trace (µm)	[WireErr]	195.8	52.0
Permissible deviation (µm)	[Fb/2]	5.0	5.0

14 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	20000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	1376

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1376

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 20000.0 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	0.0000	0.0000	1453.8212

Damage relative to the system service life (H_{att}, 1375.7 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	0.0000	0.0000	100.0000

15 Reliability calculation

15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

15.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	β	t_0	T	R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	177000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	177000	1.300e+00	9.014e+29	4.295e+30	100.0000
2	Tooth root	46959	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	46959	1.300e+00	5.823e+07	2.774e+08	0.2243

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

15.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	20000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	0.2243
Reliability R, gears subsystem (%)	[R_{subG}]	0.2243
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	1375.6850

Service life H, gears subsystem (h)	[H_{subG}]	1375.6850

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

16.2 Calculations and factors

- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with Z_β = 1/(COS(β)^0.5)
- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 714)